UNIVERSITY OF MAURITIUS
SIS 3204Y(5) – NATURAL LANGUAGE PROCESSING

MOCK PAPER 6 – Weeks 1 to 8 (Question Paper with Model Answers)
Total: 50 marks | Suggested duration: 60 minutes

Instructions: Answer ALL questions. Marks are shown beside each question. The week each question comes from is shown in the question header. Model answers are given under each question in bullet points; the mark for a point is shown in brackets, e.g. (1). Cover the answers while practising. Equivalent correct answers should receive credit.

Question 1 | Week 1 (Revision Q4) [6 marks]

Differentiate between Traditional NLP and Modern NLP.

Answer

- **Approach:** Traditional NLP uses handcrafted rules written by experts; Modern NLP uses data-driven models that learn from examples. (1)
- **Data:** Traditional uses small, task-specific datasets; Modern learns from massive amounts of text. (1)
- **Tasks:** Traditional needs separate systems for different tasks; Modern uses general-purpose models for many tasks. (1)
- **Context:** Traditional has limited context understanding; Modern understands long-range context and complex relationships. (1)
- **Ambiguity:** Traditional struggles with ambiguity and language variation; Modern handles them much better. (1)
- **Scaling:** Traditional is hard to scale as rules grow; Modern is highly scalable and improves with more data and compute. (1)
- Examples: rule-based spell/grammar checkers (Traditional); ChatGPT, Claude, Gemini (Modern).

Question 2 | Week 2 (Revision Q19) [6 marks]

Compare word-level, subword (BPE) and character-level tokenisation in the context of modern LLMs, using the text “Internationalization makes NLP powerful!”.

Answer

- **Word-level:** Internationalization, makes, NLP, powerful, ! – simple, but fails for rare or unknown words; not used in LLMs. (1)
- **Subword (BPE):** Intern, ational, ization, make, s, NLP, power, ful, ! – used in most LLMs; handles any word with meaningful pieces. (1)
- **Character-level:** I, n, t, e, r, n, a, t, i, o, n … – too many tokens, slower and less efficient. (1)
- Subword balances vocabulary size, coverage and efficiency; used in GPT, LLaMA, Gemini and Mistral. (1)
- Subword works across languages and scripts and supports code, names, slang and typos. (1)
- Fewer tokens per sentence means shorter sequences and faster training and inference. (1)

Question 3 | Week 3 (Revision Q11) [6 marks]

Explain why word embeddings are needed. In your answer, state what traditional encodings (One-Hot, Bag-of-Words, TF-IDF) miss and FOUR things word embeddings provide.

Answer

- **One-Hot:** captures identity only – no meaning, all words equally distant, sparse, poor generalisation. (1)
- **Bag-of-Words / Count:** captures presence/frequency but ignores word order and context; no similarity; high-dimensional and sparse. (1)
- **TF-IDF:** captures word importance but still has no meaning or relationships; still high-dimensional and sparse. (1)
- **Word embeddings** represent words as dense vectors of real numbers in a continuous space. (1)
- They provide **meaning** (similar words have similar vectors) and **relationships** (king − man + woman ≈ queen). (1)
- They provide **compact vectors** (50–300 dimensions), **better generalisation** and **better downstream NLP performance** (classification, QA, translation, summarisation, sentiment). (1)

Question 4 | Week 4 (Part B Q1 and Q3) [7 marks]

(a) Explain: Input → Weighted Sum → Activation → Hidden Layer(s) → Output → Prediction. (4 marks)

(b) Explain forward propagation vs backpropagation. (3 marks)

Answer

(a)

- Inputs (features) enter the network. (1)
- Each neuron computes a **weighted sum** using weights and bias. (1)
- An **activation function** is applied and hidden layers transform the information. (1)
- The output layer produces the **prediction** (or probabilities). (1)

(b)

- **Forward propagation:** information moves from input to output and produces a prediction. (1)
- The **loss** is calculated after the prediction. (1)
- **Backpropagation:** error information moves backward to compute the gradients. (1)

Question 5 | Week 5 (Sample Q3) [8 marks]

What is the vanishing gradient problem? Explain why it makes it difficult for a standard RNN to learn long-term dependencies in text.

Answer

- It happens during training with **BPTT**, when gradients are sent backwards through many time steps. (1)
- The gradients repeatedly become **smaller** and can become extremely small after many steps. (2)
- Information from earlier words has very little influence on the weight updates, so **learning from earlier time steps is reduced**. (2)
- Therefore the RNN struggles to learn relationships between words that are far apart (**long-term dependencies**). (1)
- Example: “The cat that chased the mouse was hungry”. (1)
- The model must remember “cat” when it processes “was”; if the early influence is lost, the RNN fails to learn this link. (1)

Question 6 | Week 6 (Set 1 Q3) [8 marks]

Consider the translation “The cat sat on the mat.” → “Le chat était assis sur le tapis.” Explain how attention may help when the decoder generates different French words.

Answer

- At each output step the decoder can give **different attention weights** to the English input words. (2)
- Example: when generating “chat”, “cat” is most relevant; for “était” or “assis”, “sat” becomes more relevant. (2)
- Relevant source positions receive a **higher weight**. (2)
- The weights build a **context vector** for that step, which supports the current output; the model is not forced to use the same source information for every word. (2)

Question 7 | Week 7 (Set 1 Q6) [4 marks]

a) Distinguish between training and inference in a Transformer model. (2 marks)

b) Explain the difference between a base GPT model trained for next-token prediction and an instruction-tuned LLM. (2 marks)

Answer

(a)

- **Training:** updates the parameters by optimising a loss. (1)
- **Inference:** makes predictions using fixed, already-learned parameters. (1)

(b)

- **Base GPT:** pretrained for next-token prediction. (1)
- **Instruction tuning:** uses instruction–response examples to improve how well the model follows requests. (1)

Question 8 | Week 8 (Tutorial Set B Q3) [5 marks]

A university office wants to extract the student name, module and requested action from incoming messages. The original prompt is: “Read this email and help me.”

(a) Identify two weaknesses of this prompt for automated information extraction. (2 marks)

(b) Write an improved prompt that extracts the three required fields and returns them in a consistent format. Include an instruction for missing information. (3 marks)

Answer

(a)

- It does not say **which fields** to extract. (1)
- It gives no **output format** and no rule for **missing values**. (1)

(b) Example improved prompt

- “Extract the following fields from the email: student_name, module, requested_action. Return exactly one JSON object with those three keys. If a field is not stated, use null. Do not add explanations. Email: [insert message].”
- Specifies all three fields and the extraction task. (1)
- Requires a structured, consistent output format (JSON). (1)
- Gives an explicit rule for missing fields (use null). (1)

Marks Summary

Question	Week	Source	Marks
1	1	Revision Q4	6
2	2	Revision Q19	6
3	3	Revision Q11	6
4	4	Part B Q1 and Q3	7
5	5	Sample Q3	8
6	6	Set 1 Q3	8
7	7	Set 1 Q6	4
8	8	Tutorial Set B Q3	5
		TOTAL	**50**